NLB OKR Measurement

Documentation of the app and explanation of the measures

As of 14 October 2026 · App version 1 · https://twe-nlb.github.io/NLB-OKR/

Important: The app is a demo and not a medical device. It does not make a diagnosis. The cited studies were carried out with laboratory methods (magnetic search coils, infrared oculography, large stimulus fields). The findings show which brain areas can influence a measure. They cannot be transferred directly to a single measurement with the iPad camera. Abnormal values should be clarified by a physician (neurology, neuro-ophthalmology, ENT).

Citation style: References appear in the text in square brackets: [number, section]. The number refers to the bibliography at the end. The section names the part of the study summary (abstract) the statement comes from, for example “Results” or “Conclusions”. For unstructured abstracts it says “Abstract”. The PubMed abstracts were evaluated. The full texts were not read.

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1 The optokinetic reflex in brief

When a large pattern moves across the visual field, the eyes follow it automatically. This is the optokinetic reflex (OKR). The resulting eye movement is called optokinetic nystagmus (OKN). It consists of two parts [1, Abstract]:

- **Slow phase:** The eye follows the pattern smoothly in the direction of motion. This phase reveals the most about motion processing.
- **Fast phase:** A short reset jump (a saccade) against the direction of motion brings the eye back.

Over time this produces a sawtooth. Its rhythm and height vary considerably from beat to beat, even in healthy people [8, Abstract].

Look OKN and stare OKN

There are two forms of OKN, and which one occurs depends on the instruction given to the test person:

- **Look OKN:** The person deliberately tracks individual stripes. The slow phases are long and the reset jumps are large and infrequent.
- **Stare OKN:** The person looks passively straight ahead. This is more reflexive, with shorter slow phases and lower gain [2, Purpose].

Look OKN activates considerably more cortex (mainly parietal, temporal and occipital lobes: BA7, BA19, BA21, BA39/40) than stare OKN [4, Abstract]. Stare OKN is controlled mainly by the brainstem (Mesencephalon, Pons, Medulla) and the cerebellum (Cerebellum) [16, Abstract]. The app instructs: “Look at the stripes in a relaxed way; following them is allowed.” This usually produces a mixed form with a look component. For comparable measurements the instruction must therefore always be the same.

2 The app

2.1 Technology

- **Web app:** The app is a web page. It runs in Safari on iPad or iPhone, ideally as an app on the home screen.
- **Camera:** The front camera delivers about 30 frames per second, up to 60 on some devices.
- **Eye detection:** The “MediaPipe Face Landmarker” program finds 478 facial points in every camera frame, including the iris centres and the eye corners.
- **Eye position:** The app calculates the position of the iris between the eye corners and divides it by the eye width. This keeps the value largely constant when the head or the device shifts in the image. The app averages both eyes.
- **Precedent:** Camera-based, automatically analysed OKN measurement is feasible in principle. One study used it to determine visual acuity about as well as a letter chart [39, Conclusions].
- **Data:** All measurements stay on the device. The app does not store any video. Data are backed up by exporting them as files.

2.2 Measurement workflow

1. **Person:** Create the test person under a pseudonym.
2. **Settings:** Set posture (stand or handheld), distance, device, directions, speeds, block duration, stripe width and order.
3. **Setup check:** The app checks camera, face, eyes, light, position and steadiness. Everything must be green, and the measured distance must be confirmed.
4. **Calibration:** A dot appears in turn at centre → left → centre → right → centre → top → centre → bottom → centre, for 1.8 s each. The measurement only starts when all check criteria are met (section 2.4).

5. **Measurement:** For each block the app shows a fixation cross with a direction arrow for 3 s. Then the stripe pattern runs, for 10 s by default.
6. **Result:** The app shows key values, a table of all blocks and the eye trace per block.
7. **Comparison:** When two or more measurements are ticked, the app shows a table, gain charts and an asymmetry chart.
8. **Export:** JSON (complete, re-importable) and CSV (for Excel) via the share menu: Files, AirDrop or SeaFilePro.

2.3 Settings and defaults

Setting	Default	Meaning
Posture	Stand	With “handheld”, more generous movement limits apply. The setting is saved with the measurement.
Distance eye–screen	40 cm	Converts screen points to visual angle (degrees). Please measure, don't estimate.
Device (points per inch)	iPad 132, iPad mini 163, iPhone 153	Physical size of the screen points. Needed for conversion to degrees.
Directions	← → ↑ ↓	Direction of stripe motion. Vertical stripes for horizontal motion, horizontal stripes for vertical motion.
Speeds	10, 20, 30 °/s	Stripe speed in visual angle.
Duration per block	10 s	The first second is discarded because the reflex is still building up.
Stripe width	4° per pair (0.25 cycles/°)	Width of one black plus one white stripe.
Order	all directions per speed	Spreads fatigue evenly across the directions.

2.4 Setup check and calibration

The setup check runs before calibration. Red blocks the start, yellow is allowed.

Check	Green if …
Camera	at least 20 frames per second (yellow from 12)
Face detected	in at least 90 % of the frames of the last 1.5 s
Eyes open, iris visible	palpebral fissure at least 18 % of the eye width
Light	mean image brightness between 27 % and 90 %
Face centred	midpoint between the eyes in the central area of the camera image
Head steady / device steady	variation below 3 % of the eye distance (handheld: 8 %)
Distance	Rough estimate for information; the measured distance must be confirmed

Calibration: all criteria below must be met before the stripe pattern starts.

Criterion	Limit	Purpose
Detection rate per point	≥ 60 %	Enough valid frames per gaze target
Horizontal fit (R^2)	≥ 0,8	Eye feature and gaze angle are linearly related
Horizontal separation	≥ 4 × noise	Left and right clearly distinguishable
Vertical fit (R^2)	≥ 0,7	Vertically the iris movement is smaller and harder to measure
Vertical separation	≥ 3 × noise	Up and down clearly distinguishable
Repeatability of the centre	spread ≤ 1.5° (horizontal) / 2.5° (vertical)	Fixating the centre five times gives consistent results
Head movement	≤ 25 % of the eye distance (handheld: 60 %)	Head and device stayed steady

2.5 Quality control during the measurement

- **Invalid frames:** Blink: the palpebral fissure is smaller than 55 % of the calibration value. Such frames count as invalid.
- **Shaking:** Device rotation above 20 °/s (gyroscope) or face movement in the camera image of more than 1 eye distance per second. These frames are excluded from the gain.
- **Distance correction:** If the face comes closer, the visual angle of the stripes increases. The app therefore uses the face size to convert the stripe speed to the actual distance.
- **Block quality:** “good” (gut): at least 70 % valid frames and at most 20 % shaking. “fair” (mäßig): at least 50 % and at most 40 %. Otherwise “invalid” (ungültig); the app then suggests repeating the block.

2.6 Limits of measurement accuracy

- **Frame rate:** Laboratory methods (magnetic search coil, infrared oculography) sample much faster. The app measures at about 30 frames per second. The slow phase, and thus the gain, can be captured well. Fast phases last only 20–50 ms and are therefore captured only roughly.
- **Stimulus field size:** At 40 cm the iPad covers about 30° × 22° of the visual field. Studies often use 60–70° [6, Methods]. Smaller stimulus fields lower the gain. If only the centre of the visual field is stimulated, the up–down asymmetry disappears [7, Abstract].
- **One value for both eyes:** The app averages both eyes. It cannot detect differences between the eyes, for example in internuclear ophthalmoplegia.
- **No validation data:** There are no studies yet on the repeatability of the app's measurements. How much the values vary from measurement to measurement therefore has to be determined for each person (section 5).

3 How the brain produces the OKR

The chain from the eye to the eye muscle, simplified:

1. **Retina:** The retina signals image motion, i.e. image slip across the retina (retinal slip).
2. **Pretectum (NOT/DTN):** The nucleus of the optic tract (NOT) and the dorsal terminal nucleus (DTN) in the pretectum (Mesencephalon), at the junction of thalamus and midbrain, contain direction-selective neurons. The left NOT/DTN prefers leftward motion and vice versa [17, Abstract]. In humans this region is active during horizontal and vertical OKN [15, Abstract].
3. **Cerebral cortex:** MT/V5 and MST, at the junction of the temporal and occipital lobes, process motion (BA19 and partly BA37). They send information to NOT/DTN and to the pontine nuclei [19, Abstract]. During OKN, visual motion areas and parietal and frontal eye fields are active, with emphasis on the right hemisphere [13, Abstract].
4. **Vestibular nuclei (BA40 and parts of BA21/22) and velocity storage (Medulla, Pons, Cerebellum):** The vestibular nuclei combine visual and balance signals to estimate self-rotation [20, Abstract]. A storage mechanism (velocity storage) sustains eye velocity. It also produces the after-nystagmus in darkness [21, Abstract].
5. **Cerebellum (Cerebellum):** Flocculus, uvula and oculomotor vermis are active during OKN [15, Abstract; 16, Abstract]. Damage to the uvula and pyramis is associated with reduced slow-phase gain [22, Abstract].
6. **Saccade generators (BA7, BA40, Pons, Mesencephalon):** The fast phases arise in two centres [27, Abstract]:
    - horizontal: in the paramedian pontine reticular formation (PPRF) of the pons (Pons) [26, Abstract],
    - vertical: in the rostral interstitial nucleus of the medial longitudinal fasciculus (riMLF) in the midbrain (Mesencephalon) [24, Abstract].

fMRI shows activation of the PPRF during horizontal OKN and of the riMLF during vertical OKN [15, Abstract].

7. **Eye-muscle nuclei (Pons, Mesencephalon):** Nuclei III (oculomotor nerve), IV (trochlear nerve) and VI (abducens nerve) control the extraocular muscles [28, Abstract].

https://www.youtube.com/watch?v=WlTFA99sSlI&t=11s

3.1 Eye muscles by direction of movement

The following assignment is anatomical textbook knowledge. The cited papers document the nuclei and connections.

Eye movement	Muscles	Nerve / nucleus	Mediated by
to the right (both eyes)	right lateral rectus + left medial rectus	right VI (Pons), left III (Mesencephalon)	Abducens interneurons (Pons) via the medial longitudinal fasciculus (MLF) [29, Abstract]
to the left (both eyes)	left lateral rectus + right medial rectus	left VI (Pons), right III (Mesencephalon)	as above, mirrored
upward	superior rectus + inferior oblique	III (Mesencephalon)	riMLF, interstitial nucleus of Cajal (Mesencephalon) [29, Abstract]
downward	inferior rectus + superior oblique	III and IV (Mesencephalon)	riMLF (Mesencephalon) [25, Abstract]

When the stripes move to the right, both eyes move to the right in the slow phase. The reset jump goes to the left. Horizontal saccades are generated by the PPRF (Pons) on the side towards which they move [26, Abstract].

4 The measures

Each measure is described with: definition, calculation in the app, normal values, meaning, brain areas and eye muscles involved, typical changes in studies, and notes on comparing several measurements.

4.1 Gain (slow-phase gain)

Definition: Eye velocity in the slow phase divided by the stripe speed. 1.0 means the eye moves exactly as fast as the stripes. 0.5 means half as fast.

How the app calculates it: Eye position in degrees (via calibration) → frame-to-frame velocity → remove fast phases and shaking → median of the remaining velocities ÷ distance-corrected stripe speed. For every block, i.e. every direction and every speed.

Normal values and influencing factors

- **Healthy adults:** The gain was always below 0.85 and decreased with increasing speed [5, Abstract].
- **Age:** The gain rises in the first years of life, stays stable until about age 50 and then falls by 6–18 % [9, Results]. A study of 250 healthy people aged 18 to 70 also found a decline with age [12, Abstract]. For tracking of a large stripe pattern, gain and precision declined progressively with increasing age [10, Abstract].
- **Stimulus:** The gain is highest at low speed and with coarse stripes [6, Results]. Another study, however, found higher eye velocities with finer stripes [8, Abstract]. Therefore never change the stripe width between measurements you want to compare.
- **Binocular:** Binocular viewing gives a higher gain than monocular viewing [5, Abstract].
- **Instruction:** Look OKN has a higher gain than stare OKN [2, Purpose].

Meaning

- **General:** The gain shows how well the brain detects image motion and converts it into an equally fast eye movement. The goal is a stable retinal image.
- **For the app:** Because of the smaller stimulus field, somewhat lower values than in laboratory studies are to be expected (section 2.6).

Brain areas and eye muscles involved

- **Input:** Retina → NOT/DTN (Mesencephalon) [17, Abstract] and cortical motion areas MT/MST (BA19/37) [19, Abstract].
- **Processing:** Vestibular nuclei (Medulla, Pons) and velocity storage (Medulla, Pons, Cerebellum) [20, Abstract], cerebellum (Cerebellum: uvula, pyramis, flocculus) [22, Abstract].
- **Execution:** horizontal: lateral and medial rectus (III, VI); vertical: superior/inferior rectus and the oblique muscles (III, IV).

Typical changes in studies

- **Cerebellum (Cerebellum):** Patients with damage to the uvula and pyramis had reduced gain of the slow OKN phase [22, Abstract].
- **Cerebellar degeneration (Cerebellum):** Horizontal optokinetic stimuli sometimes triggered an additional rotation of the eye around the line of sight. The app does not measure this component [23, Abstract].
- **Multiple sclerosis:** Slow-phase velocity during the rise time was reduced [36, Abstract].
- **PSP:** Vertical gain was reduced in progressive supranuclear palsy, but mostly normal in Parkinson's disease [30, Results].
- **Age:** In people over 75, OKN gain declined over 9–12 years. The decline was related to gait and balance scores [11, Abstract].

Comparing two or more measurements

- **Same person:** If the gain falls evenly in all directions, the most common causes are attention, fatigue, light or a different distance. Only if it is confirmed on repetition should a general change in the slow tracking system be considered (cerebellum/Cerebellum, age).
- **Different people:** Only compare people of similar age [9, Results], with the same device, distance and posture.
- **Display in the app:** The table shows the gain per direction and speed and the difference from the oldest measurement, absolute and in %. The “Gain je Richtung” (gain per direction) chart shows one line per measurement.

4.2 Gain across speeds

Definition: How much the gain drops from 10 through 20 to 30 °/s, i.e. the slope of the lines in the “Gain je Richtung” chart.

How the app calculates it: Not a separate value. The app shows the curve in the result (block table) and in the comparison (four charts).

Normal values and influencing factors

- **Normal:** The gain decreases with increasing speed [5, Abstract]. Vertically it drops more steeply than horizontally [5, Abstract].
- **Cause:** Above a certain speed the tracking system can no longer keep up. The gain “saturates”.

Meaning

- **General:** The curve shows the performance reserve of the tracking system. A steeper drop than before, or than in comparison persons, suggests that fast stimuli are processed less well.

Brain areas and eye muscles involved

- **Involved:** Mainly the fast, direct tracking pathway via cortical motion areas (BA19/37), pontine nuclei (Pons) and flocculus (Cerebellum) [18, Abstract].
- **Measurement in humans:** fMRI found speed-dependent activation in the pons (Pons: NRTP, PPRF), in the uvula, flocculus and cerebellar tonsils (Cerebellum) [16, Abstract].

Comparing two or more measurements

- **Same person:** If the gain drops more than before only at 30 °/s, this is more likely a change in tracking speed than in motion perception as a whole.
- **How to read it:** Parallel-shifted lines mean the gain has changed overall. Lines with different slopes mean the speed dependence has changed.

4.3 Slow-phase velocity

Definition: Eye velocity in degrees per second during tracking (in the CSV: “Langsame Phase °/s”).

How the app calculates it: Median of the frame-to-frame velocity, excluding reset jumps and shaking. The gain is this value divided by the stripe speed.

Normal values and influencing factors

- **Normal:** At 10 °/s it is just below 10 °/s. At higher speeds it falls increasingly behind [5, Abstract].

Meaning

- **General:** The absolute value is easier to compare across studies with different stripe speeds. For comparisons within the app the gain is more suitable.

Brain areas and eye muscles involved

- **Involved:** as for the gain (section 4.1).

Typical changes in studies

- **Concussion:** After concussion, slow phases were faster and visual gain higher for rotational stimuli around the line of sight [38, Results]. The app does not measure this type of stimulus.

Comparing two or more measurements

- **Comparison:** Only meaningful at the same stripe speed; otherwise use the gain.

4.4 Beat frequency

Definition: Number of reset jumps (fast phases) per second.

How the app calculates it: Reset jumps are detected when the eye jumps against the direction of motion faster than max(25 °/s, 1.2 × stripe speed) and covers at least 0.5°. Count ÷ analysed time.

Normal values and influencing factors

- **Typical:** Stare OKN has frequent small reset jumps, look OKN infrequent large ones [3, Abstract].

- **Relationship:** Frequency and amplitude depend on the slow-phase velocity and vary strongly from beat to beat [8, Abstract].

Meaning

- **General:** The frequency mainly shows how the person looks (strategy, attention) and how well the fast phases are triggered.
- **Limit of the app:** At about 30 frames per second, very small reset jumps in quick succession can merge or be missed.

Brain areas and eye muscles involved

- **Horizontal reset jumps:** Burst neurons of the PPRF in the pons (Pons) [26, Abstract; 27, Abstract].
- **Vertical reset jumps:** riMLF in the midbrain (Mesencephalon). If it is destroyed on both sides, fast vertical eye movements are impaired [24, Abstract].
- **Execution:** the same muscles as in the slow phase, but in the opposite direction (section 3.1).

Typical changes in studies

- **Neglect:** In patients with left-sided neglect after damage to the right hemisphere (neglect typically arises from damage to the right inferior parietal lobe, BA39/40), velocity, amplitude and frequency of the phases with rightward slow movement were reduced [35, Abstract].
- **Multiple sclerosis:** In one study beat frequency was unchanged, but velocity and amplitude were reduced [36, Abstract].

Comparing two or more measurements

- **Same person:** If only the frequency changes and the gain stays the same, the person was usually looking differently (more “look” or more “stare”). This does not indicate altered brain function.
- **More significant:** If reset jumps are missing or less frequent in one direction only while the gain is preserved, the saccade system for that direction is affected, not the tracking system.

4.5 Amplitude

Definition: Size of the reset jumps in degrees, as the median of all detected reset jumps in a block.

How the app calculates it: Largest drop in position within a detected reset jump, in degrees via calibration.

Normal values and influencing factors

- **Normal:** Over 95 % of vertical reset jumps are smaller than 10° [6, Results].
- **Instruction:** Look OKN has large amplitudes, stare OKN small ones [3, Abstract].

Meaning

- **General:** The amplitude shows how far the eye follows the pattern before jumping back. Together with the frequency it describes the shape of the sawtooth.

Brain areas and eye muscles involved

- **Involved:** Saccade generators (PPRF: Pons; riMLF: Mesencephalon) and eye-muscle nuclei (III, IV: Mesencephalon; VI: Pons) [27, Abstract].
- **Beating field:** During nystagmus the mean gaze position shifts by about 2.6° towards the fast phases. This also changes the activation of the visual cortex (BA17/18) [14, Abstract].

Typical changes in studies

- **Hemisphere:** With damage to the entire left hemisphere, OKN towards the opposite side had a low amplitude and reduced slow-phase velocity [33, Abstract].
- **Multiple sclerosis:** Amplitude was reduced [36, Abstract].
- **PSP:** Vertical reset jumps were slowed and combined with small horizontal saccades [30, Results].

Comparing two or more measurements

- **Same person:** As with frequency, first consider a different gaze strategy. What matters is a direction in which the amplitude alone decreases, together with a gain difference.

4.6 Right/left asymmetry

Definition: (gain right − gain left) ÷ mean of both × 100 %. Positive: stripes moving to the right are tracked better.

How the app calculates it: Mean of the valid blocks per direction across all speeds, then the formula above.

Normal values and influencing factors

- **Normal:** Horizontally there is no group asymmetry in healthy people [5, Abstract; 2, Results]. According to a review, horizontal OKN is symmetrical in healthy adults [1, Abstract].
- **Infants:** With monocular testing there is an asymmetry in the first months of life. Only from about 11 months is OKN symmetrical in all infants [9, Results].

Meaning

- **General:** A clear and repeatable asymmetry is the most important clinical finding in horizontal OKN. It points to a one-sided disturbance of motion processing or gaze control.

Brain areas and eye muscles involved

- **Pretectum (Mesencephalon):** After damage to NOT/DTN, OKN with slow phases towards the damaged side was reduced. The opposite direction was unaffected [18, Abstract] (study in monkeys).
- **Cerebral cortex:** After damage in the region of the frontal eye field (BA8, transition to BA6) and the posterior parietal lobe (BA7, BA39/40), tracking towards the side of the damage was slowed; with optokinetic stimulation in 8 of 13 cases with parietal lesions [34, Abstract].
- **Hemisphere:** With damage to the whole left hemisphere, OKN towards the opposite side was reduced. With purely parietal lesions (BA7, BA39/40) there was no directional preponderance [33, Abstract].
- **Muscles:** If the asymmetry is muscular, the lateral rectus (VI) or medial rectus (III) of one side is affected. This cannot be distinguished using the average of both eyes.

Typical changes in studies

- **Internuclear ophthalmoplegia:** With a lesion of the medial longitudinal fasciculus (MLF; Pons/Mesencephalon), adduction of the eye on the side of the lesion is impaired [29, Abstract]. Tracking and OKN were studied in this condition [32, Abstract]. The app cannot detect this because it averages both eyes.

Comparing two or more measurements

- **Same person:** An asymmetry that appears for the first time or increases clearly is the most meaningful comparison finding. It should be confirmed by repeating with the same setup.
- **Different people:** How large the normal spread of horizontal asymmetry is with the app has yet to be shown by a collection of measurements from healthy people.

4.7 Up/down asymmetry

Definition: (gain up − gain down) ÷ mean of both × 100 %. Positive: upward-moving stripes are tracked better.

How the app calculates it: as in section 4.6, with the directions ↑ and ↓.

Normal values and influencing factors

- **Mixed evidence:** Several studies found an upward advantage, on average about 0.15 gain [5, Abstract]; likewise [6, Results] and at 30–70 °/s [7, Abstract]. Others found no vertical asymmetry at group level [2, Results].
- **Individual trait:** Vertical asymmetries differ between individuals but remain constant within the same person [2, Conclusions].
- **Stimulus field:** The up–down asymmetry depends on the periphery of the visual field. With purely central stimulation it disappeared [7, Abstract]. On the iPad it is therefore probably smaller than in the laboratory.

Meaning

- **General:** Because of the large individual differences, the vertical value is suited mainly to tracking change within the same person, less to comparisons between people.

Brain areas and eye muscles involved

- **Vertical gaze centres:** riMLF, interstitial nucleus of Cajal and posterior commissure in the midbrain (Mesencephalon) [29, Abstract; 15, Abstract].
- **Muscles:** upward: superior rectus + inferior oblique (III); downward: inferior rectus (III) + superior oblique (IV).

Typical changes in studies

- **Parkinson's disease:** Vertical OKN was strongly asymmetrical, with reduced gain for downward-moving stimuli, more pronounced than in controls [31, Abstract].
- **PSP:** Vertical gain was reduced and vertical reset jumps were slowed [30, Results]. An isolated loss of vertical eye movements points to the midbrain (Mesencephalon) [29, Abstract].
- **riMLF (Mesencephalon):** Even a one-sided lesion of the riMLF can cause an upgaze palsy or a combined up- and downgaze palsy [25, Abstract].

Comparing two or more measurements

- **Same person:** Because the personal vertical asymmetry is stable [2, Conclusions], a change relative to one's own earlier values is more meaningful than the value itself.
- **Different people:** Differences between people in the vertical value are often normal.

4.8 Technical values

These values describe measurement quality, not brain function:

Value	Meaning	What to do if values are poor
Eyes detected (%)	Share of valid frames in the block	Light, glasses, camera at eye level
Shaking (%)	Share of frames with device or head movement	Use a stand, support the head
eff. °/s	Stripe speed converted to the actual distance	If it differs strongly from the set value, the distance was different from the setting
Quality	good / fair / invalid (section 2.5)	Repeat invalid blocks
Calibration R^2 / separation	Quality of the conversion to degrees	Recalibrate if values are borderline

4.9 Not measured: after-nystagmus

When the stripe motion stops in darkness, the nystagmus briefly continues (optokinetic after-nystagmus, OKAN). It reflects velocity storage in the vestibular nuclei (Medulla, Pons) and cerebellum (Cerebellum) [21, Abstract]. After damage to NOT/DTN (Mesencephalon) it was reduced or absent [18, Abstract]. In MS its time constant was shortened [36, Abstract]. In patients with persistent symptoms after concussion it was prolonged [37, Abstract]. The app does not measure OKAN yet. It would be a possible extension: switch the screen to black at the end of a block and keep measuring.

5 Comparing two or more measurements

5.1 What the app shows

- **Reference:** The oldest selected measurement is the reference.
- **“Zusammenfassung” (summary) table:** Overall, horizontal and vertical gain, right/left and up/down asymmetry, beat frequency and amplitude, each with the difference from the reference (absolute; for gain also in %).
- **“Gain je Richtung und Geschwindigkeit” table:** Gain per direction and speed, with the difference.
- **“Gain je Richtung” charts:** Four charts, one per direction, with one line per measurement across 10, 20 and 30 °/s.
- **“Asymmetrie” chart:** Bars for horizontal and vertical asymmetry, one bar per measurement.

5.2 When is a difference meaningful?

There are no studies yet on the measurement accuracy of the app. Until then, the following practical procedure applies. It is a recommendation, not a literature value:

1. **Baseline:** Take 2–3 measurements per person on the same day at the start. How much they differ from each other is that person's own measurement variability.
2. **Real change:** A later difference only counts as real if it is clearly larger than this variability and is confirmed on repetition.
3. **Pattern over single value:** A difference affecting only one direction is more meaningful than an even change across all directions. Even changes often have external causes such as fatigue, light or distance.
4. **Quality:** Check quality and shaking of the compared blocks. A “fair” block can create a spurious difference.

5.3 Patterns of change and possible meaning

The table shows which patterns were associated with which brain areas in studies. It is a guide, not a diagnostic manual.

Pattern in the comparison	Possible meaning	Brain area / eye muscle	Source
Gain evenly lower in all directions	Attention, fatigue, setup; age; general weakness of the tracking system	Cerebellum (Cerebellum: uvula, pyramis); age-related	[22, Abstract], [9, Results], [11, Abstract]
One horizontal direction lower (new R/L asymmetry)	one-sided disturbance of motion processing or tracking control	NOT/DTN (Mesencephalon) on the side towards which the weaker slow phase points (animal study); frontal eye field (BA8/6) / posterior parietal lobe (BA7, BA39/40);	[18, Abstract], [34, Abstract], [33, Abstract]

Pattern in the comparison	Possible meaning	Brain area / eye muscle	Source
		hemisphere	
Vertical downward worse than before	altered vertical tracking	Basal ganglia influence (subcortical; substantia nigra in the Mesencephalon) (Parkinson's finding)	[31, Abstract]
Lower vertical gain, reset jumps rare or oblique	impaired vertical saccade system	Midbrain (Mesencephalon): riMLF; muscles III/IV	[30, Conclusions], [24, Abstract]
Horizontal reset jumps missing with preserved gain	impaired horizontal saccade generation	Pons (Pons): PPRF	[26, Abstract], [29, Abstract]
Only frequency/amplitude differ, gain the same	different gaze strategy (look/stare), attention	cortical eye fields (BA8/6, BA7/40) in look OKN	[3, Abstract], [4, Abstract]
Gain drops more steeply at 30 °/s	lower reserve of the fast tracking system	Pontine nuclei (Pons), flocculus, uvula (Cerebellum)	[16, Abstract]
Gain higher than before	e.g. look instead of stare instruction; after concussion a higher visual gain has been described (different stimulus type)	subcortical gaze stabilisation	[2, Purpose], [38, Conclusions]

5.4 Comparing the same person over time

- **Strength:** This is the most meaningful use. Individual traits, such as vertical asymmetry, remain the same within one person [2, Conclusions].
- **Age effect:** From about age 50 the gain slowly declines [9, Results]. In the very old it declined over years, together with gait and balance [11, Abstract]. A slight decline over years is therefore normal.
- **Requirements:** Same device, same distance, same posture, same instruction, similar time of day, same light.

5.5 Comparing different people

- **Age:** Only compare people of similar age [9, Results; 12, Abstract].
- **Asymmetries:** Vertical asymmetries differ between individuals and are a personal trait [2, Conclusions]. Horizontal asymmetries are rare in healthy people [1, Abstract].
- **Posture:** Do not compare “handheld” measurements directly with “stand” measurements. The app marks handheld measurements in the comparison.

6 Recommendations for good measurements

- **Setup:** iPad in landscape in a stand, camera at eye level, distance measured with a tape measure.
- **Light:** even, from the front, no backlight from a window.
- **Glasses:** reflective lenses make detection harder. For comparisons always do the same (with or without).
- **Instruction:** word for word the same every time, for example: “Look at the stripes in a relaxed way.”
- **Settings:** never change stripe width, speeds or block duration within a series of measurements.
- **Repetition:** repeat invalid blocks immediately. Take several baseline measurements at the start.
- **Backup:** export after every session, for example via AirDrop or to SeaFile.

7 Glossary

Term	Explanation
OKR / OKN	optokinetic reflex / optokinetic nystagmus: tracking movement of the eyes in response to a moving pattern
Slow phase	Tracking movement in the direction of motion
Fast phase	Reset jump (saccade) against the direction of motion
Gain	Eye velocity ÷ stripe speed
Look / stare	active tracking / passive looking
NOT / DTN	nucleus of the optic tract / dorsal terminal nucleus: entry station of the OKR in the pretectum (Mesencephalon)
MT/V5, MST	Motion areas of the cerebral cortex (BA19/37)
PPRF	paramedian pontine reticular formation: centre for horizontal saccades (Pons)
riMLF	rostral interstitial nucleus of the medial longitudinal fasciculus: centre for vertical saccades (midbrain, Mesencephalon)
MLF	medial longitudinal fasciculus: connects the abducens (Pons) and oculomotor nuclei (Mesencephalon)
Flocculus, uvula, vermis	Parts of the cerebellum (Cerebellum) that fine-tune eye movements
Velocity storage	Velocity storage in the vestibular nuclei (Medulla, Pons) and cerebellum (Cerebellum)
OKAN	optokinetic after-nystagmus: continuation of the nystagmus in darkness
PSP	progressive supranuclear palsy
INO	internuclear ophthalmoplegia: weakness of adduction with an MLF lesion

8 Before-and-after measurement with training

Measuring the same person before and after eye training is the most meaningful use of the app. This applies to children and adults. Each person is their own reference, and the large differences between people do not matter. For a change to mean something, the following points are important.

8.1 Know the personal variability

Values vary from day to day even without training, especially in children. How much this applies to the app has not yet been studied.

1. **Baseline measurements:** Before the training, take 2–3 baseline measurements on different days, e.g. within 1–2 weeks. These measurements show the person's normal variability.
2. **Threshold:** A change after the training only counts if it is clearly larger than this variability. Practical rule of thumb: more than twice the range of the baseline measurements. This is a recommendation, not a literature value.
3. **Follow-up measurements:** Also take 2 measurements after the training. A single measurement can be an outlier.

Time point	Measurements	Purpose
Weeks 1–2, before the training	2–3, on different days	determine personal baseline variability
Training phase	optional: 1 interim measurement	see the trend
After the training	2, on different days	test the change against the baseline variability
A few weeks later (optional)	1–2	check whether a change persists

8.2 Keep everything the same

- **Setup:** Device, distance, stand, light and time of day.
- **Settings:** Stripe width, speeds, block duration and order.

- **Instruction:** word for word the same. In children this is the biggest confounder: a child who pays more attention the second time has a higher gain without anything having changed in the brain. Look OKN has a higher gain than stare OKN [2, Purpose].

8.3 What the measurement cannot prove

If a value changes, there can be many reasons, not just the training:

- **Familiarity with the test:** The person knows the procedure and is more relaxed.
- **Attention and motivation:** Form on the day, fatigue, interest.
- **Normal development in children:** OKN gain rises in the first years of life and then stays stable until about age 50 [9, Results]. In 7- to 14-year-olds OKN was described as stable [49, Conclusion]. Over a few weeks the developmental effect is therefore likely small.
- **Regression to the mean:** Someone who happened to do poorly in the first measurement often does better the next time, even without training.
- **Training transfer:** If the eye training resembles the stripe pattern, the test task itself may mainly be practised. A general improvement is then not demonstrated.

Important: The measurement can show that something has changed in this person. On its own it does not prove that the training was the cause. That would require a comparison group without training or a single-case design: training starts at different times for several people. If the values improve only after training begins in each case, that is much stronger evidence for the training.

8.4 Which measures are most meaningful

- **Most suitable:** Changes in asymmetry (right/left, up/down) and in gain at 30 °/s. They depend less on attention than the overall gain.
- **Less suitable:** Beat frequency and amplitude often change simply because the person looks differently [3, Abstract]. They are less suitable as outcome measures.
- **Quality:** Check “Augen erkannt” (eyes detected) and “Wackeln” (shaking) for every measurement. Only compare blocks of good quality.

8.5 Framework

- **Consent and data protection:** For children, parental consent is required. Use pseudonyms only. The data stay on the device.
- **Wording:** Word results cautiously, especially in publications. Appropriate: “In person X, the gain rose after the training beyond their own baseline variability.” Not appropriate: “The training improves OKN.”

9 Evidence: neurodevelopmental disorders

PubMed was searched for OKN in connection with ADHD, autism, dyslexia, tics/Tourette, coordination disorders and genetic syndromes (as of October 2026). The abstracts were evaluated. The figures “n = … vs. …” give the size of the patient and comparison groups.

Key message: None of these disorders has an established OKN feature. The findings come from few, small and partly old studies. The app therefore cannot assign anyone to a diagnosis. It is suitable for tracking change within the same person (chapter 8).

Disorder	Evidence	Findings
Dyslexia	weak to moderate	lower OKN gain; larger difference between the two eyes
Autism	weak, contradictory	lower gain with more autistic traits; brainstem reflexes

Disorder	Evidence	Findings
		intact
ADHD	very weak	OKN matures more slowly
Tics / Tourette	case reports only	OKN once normal, once interrupted by saccades
Down syndrome (for comparison)	moderate	gain reduced depending on speed, fewer beats

9.1 Dyslexia

- **Lower gain:** Children with dyslexia had lower gain in pursuit and OKN (n = 30 vs. 20, video-oculography) [40, Results].
- **Difference between the eyes:** The OKR was reduced only in the right eye, so the difference between the eyes was larger. Saccades were clearly slower and less accurate (n = 40 vs. 40) [41, Results]. The authors stress that the association is not shown to be causal [41, Conclusions].
- **Unsteady gaze:** Against a moving optokinetic background, gaze became more unstable than in controls [42, Abstract].
- **Meta-analysis:** A meta-analysis found mainly longer fixations and fewer saccades, no OKN differences [43, Conclusion].
- **Caution:** Studies on a "cerebellar-vestibular dysfunction" come from a single author. They themselves call for independent confirmation [44, Abstract].

9.2 Autism

- **Gain and autistic traits:** In healthy adults, the higher the autism score (Autism-spectrum Quotient), the lower the slow-phase gain. There was no relationship for the fast phases [45, Abstract]. The participants did not have an autism diagnosis.
- **Unusual behaviour during the test:** In 31 of 34 children with autism the OKN was atypical: delayed onset, short duration, looking away [46, Abstract]. This mainly concerns attention and gaze avoidance, not the reflex itself.
- **No difference:** A larger vision screening (n = 51 vs. 44) reported no clear OKN differences. Only convergence stood out [47, Abstract].
- **Brainstem and cerebellum intact:** Reflexes of the balance system and saccade velocity were normal; pursuit showed only a trend towards lower gain (n = 79 vs. 62). The authors locate the differences in the cerebral cortex, not in the brainstem (Pons, Medulla) or cerebellum (Cerebellum) [48, Abstract].

9.3 ADHD

- **Slower maturation:** In 148 vs. 148 boys (aged 7–14), slow-phase OKN velocity was lower and its development delayed [49, Results]. Only the abstract is in English, and some of its figures seem implausible.
- **Biggest problem:** Inattention lowers the gain regardless of any brain difference. In ADHD the two can hardly be separated.

9.4 Tics and Tourette syndrome

- **Case report 1:** A girl with eye tics had normal OKN [50, Abstract].
- **Case report 2:** In a patient with Tourette syndrome, intruding saccades interrupted pursuit and OKN [51, Abstract].
- **Research focus:** Studies on Tourette syndrome mainly examine saccades and fixation, for example fewer and smaller microsaccades [52, Abstract]. A review describes different eye-movement patterns in autism, ADHD and Tourette, mainly for saccades [53, Recent Findings].

9.5 For comparison: Down syndrome

- **Clearest finding:** OKN gain dropped more steeply with increasing speed than in controls, and there were fewer nystagmus beats (n = 32 vs. 32) [54, Abstract].
- **Confirmation:** All children examined had abnormal gain values in at least one test. The cerebellum (Cerebellum) is suspected as the cause [55, Conclusions].

9.6 Implications for the app

1. **No diagnosis:** No study shows an OKN feature reliable enough to assign a diagnosis.
2. **Group comparisons:** Possible, but only with age-matched controls and the same instruction. In children, attention strongly determines the result.
3. **Difference between the eyes:** The dyslexia finding of a difference between the eyes [41, Results] cannot be measured with the current app because it averages both eyes.
4. **Autism:** For the autism question, the gain under a stare instruction (“just look straight ahead, don't follow”) would be the appropriate measure [45, Abstract].

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All papers are listed in PubMed (https://pubmed.ncbi.nlm.nih.gov/<PMID>). The section references in the text refer to the PubMed abstract.

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